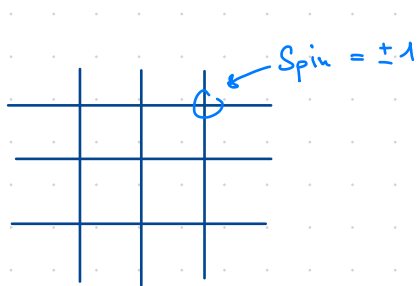
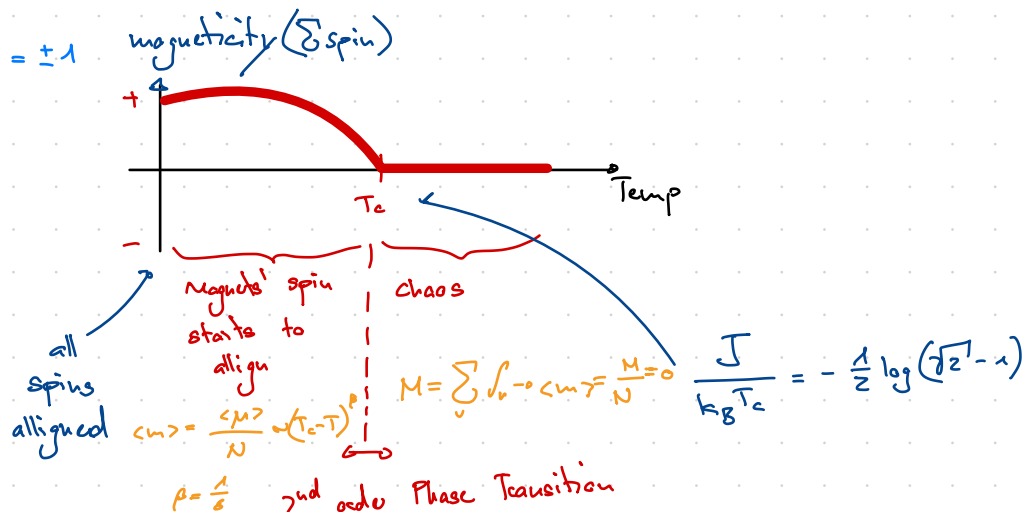


2D Ising



Example: Ferrous Magnet



Hamiltonian of System:

$$\text{Energy } H = -J \sum_{\mu, \nu} s_{\mu} s_{\nu}$$

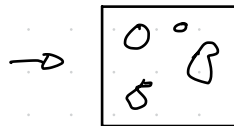
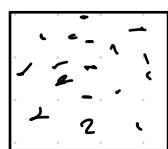
Interaction

Energy

↳ $J > 0 \rightarrow$ ferromagnetic System $\rightarrow$ permanent Magnetism

↳ $J < 0 \rightarrow$ antiferromagnet

$\rightarrow$ No analytic solution for $2 < \text{dimension}$



$$\langle M \rangle = \int_{C_i} P(C_i) M(C_i) dC_i$$

Monte Carlo Simulation

$$C_1 \rightarrow C_2 \rightarrow \dots \rightarrow C_N$$

$\underbrace{\hspace{10em}}_{\langle M \rangle}$

Markov Process

$$C_{S_1} \rightarrow C_{S_2} \rightarrow \dots \rightarrow C_{S-N}$$

$$J = k_b = 1$$

Implement:

1. Pick Random spin to flip or not to flip

2. $\Delta E = ?$ from Neighbors

3. $\Delta E < 0 \rightarrow$ accept and flip

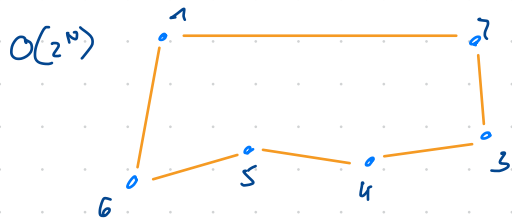
↳ $\Delta E > 0 \rightarrow$ pick random $0 \leq r \leq 1$

↳ if $r < e^{-\Delta E / T}$

Flip

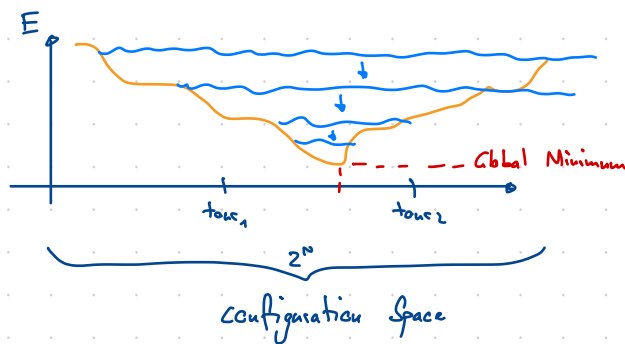
$$\Delta E = -J \sum_{\nu} s_{\nu} s_{\mu}$$

Travelling Salesman Problem



→ NP-Complete
 ↳ Non-deterministic Polynomial
 ↳ Class of unsolved problems

$$E_{\text{tours}} = \text{Length}_{\text{tour}} = \sum_{i=1}^N \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} \quad \begin{matrix} x_0 = x_n \\ y_0 = y_n \end{matrix}$$



$$T_1 = (1-\epsilon)T_0$$

$\epsilon = 0.01$

$$C_i = \{1, 2, 3, 4, 5, 6\} \rightarrow \text{Optimum}$$

$$C_{i+1} = \{1, 4, 3, 2, 5, 6\}$$

Detailed Balance → We can get every condition

$$P(E_i, E_{i+1}, T) = \begin{cases} 1 & E_{i+1} < E \\ e^{\frac{(E_i - E_{i+1})}{T}} > \text{Random} > 0 & E_{i+1} > E \end{cases}$$

$$C_i \xrightarrow{\text{any}} C_j \text{ in } O(N^2) \text{ moves}$$

A
 swap 2 random "Cities"

$$C_i = \{1, 2, 3, 4, 5, 6\}$$

$$C_{i+1} = \{1, 5, 4, 3, 2, 6\}$$

Detailed Balance
 → get to every other config in $O(N^2)$

→ Experience shows that **A** and **B** should be combined

→ Initial energy level with:

